Curriculum vitae - 25/01/2026

Personal Information and Contact

Name: Oscar Wallnoefer
Birth: 09/02/1999
Nationality: Italian, Austrian
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Address: Department of Biological, Geological and Environmental Sciences, Via Selmi 3, 40126, Bologna, Italy
GitHub: https://github.com/oscarwallnoefer

Current role

09/2025 - now **Visiting student**
Arctic University of Norway
PI: Prof. Bastian Fromm
Project title: Annotation and characterization of microRNAs in Gastrotricha and Tardigrada

2023 - now **PhD student - Earth, Life and Environmental Sciences**
University of Bologna
PI(s): Prof. Marco Passamonti, Prof. Federico Plazzi
Title (provisional): Phylogenomic Inference and Mitonuclear Coevolution in Metazoa

Education

2021 - 2023 **Master's Degree in Biodiversity and Evolution**
University of Bologna
Final grade: 110/110 cum laude
Thesis: "*The phylogenetic analysis based on nuclear OXPHOS genes reveals clues of mitonuclear coevolution in Squamata*" - Supervisor: Prof. Federico Plazzi

2018 - 2021 **Bachelor's Degree in Biology**
University of Bologna, 2018 – 2021
Final grade: 102/110
Thesis: "*Population density of the solitary azooxanthellate coral Caryophyllia inornata along an environmental gradient generated by CO_2 vent*" - Supervisor: Prof. Stefano Goffredo

PhD Projects

1. Spiralia Phylogenomics

Designed and conducted the first phylogenomic analysis of the Gastrotricha phylum (under review). Currently conducting Spiralia phylogenomics, with particular focus on Rouphozoa (ongoing).

2. Mitonuclear coevolution and its impact on phylogenetic inference

Designed and conducted two projects (one published, one manuscript) investigating the intertwined coevolution between mitochondrial and nuclear proteins and the effects on phylogenetic signal in Squamata.

3. Small non-coding RNAs across Metazoa

Annotating the microRNA complement in microscopic invertebrates such as gastrotrichs (ongoing) and tardigrades (ongoing). Investigating mitochondrial sncRNAs in the Manila clam *Ruditapes philippinarum* (ongoing).

Additional projects as Co-author:

4. Gastrotricha mitogenomes and phylogenetics (published)
5. Mitonuclear coevolution in Aves (ongoing)

Publications

1. (**manuscript**) Wallnoefer O., Martini M., Galletti G., Gabriele N., Plazzi F., Passamonti M. A Network-based Characterization of Mitonuclear Coevolution in Squamata.
2. (**under revision**) Wallnoefer O., Kosakyan A., Cesaretti A., Saponi F., Gammuto L., Serra V., Petroni G., Plazzi F., Todaro M.A. A Phylogenomic Framework of Gastrotricha Evolutionary Relationships. Available at SSRN: http://dx.doi.org/10.2139/ssrn.5818579
3. (**published**) Kosakyan, A., Gammuto, L., Cesaretti, A., Saponi, F., Serra, V., Petroni, G., Macher, J.N., Wallnoefer, O., Plazzi, F. and Todaro, M.A., 2026. From rigid order to radical variation: mitogenome evolution in the main lineages of a lesser-known animal phylum (Gastrotricha). *Genome Biology and Evolution*, p.evag001.
4. (**published**) Wallnoefer, O., Formaggioni, A., Plazzi, F. and Passamonti, M., 2025. Convergent evolution in nuclear and mitochondrial OXPHOS subunits underlies the phylogenetic discordance in deep lineages of Squamata. *Molecular Phylogenetics and Evolution*, 208, pp.1-11.
5. (**published**) Cassarino, C., Mancuso, A., Prada, F., Sani, T., Dall'Ara, S., Wallnoefer, O., Marchini, C., Tassi, F., Campanelli, A., Marini, M. and Hammel, J.U., 2025. Newly discovered CO_2 (carbon dioxide) vent cave drives r-strategy shift in a Mediterranean aphotoendosymbiotic coral. *Science of the Total Environment*, 1005, p.180858.

Technical Skills

Scripting	**Bioinformatics**	**Wet Lab**
Bash	Phylogenomics, phylogenetics	DNA/RNA extraction
R	WGS/RNAseq data analysis	PCR
Git	genome assembly	qPCR
Python	genomic networks	TapeStation
	differential expression analysis	microRNA sequencing
	microRNA annotation	Small-RNAseq library preparation

Conferences

- (talk) Evolution 2024 - Virtual Conference
- (talk) Italian Zoological Union 2024 - Pisa (IT)
- (talk) Evolution 2025 - Virtual Conference
- (talk) Willi Hennig Society 2025 - Hong Kong
- (talk) SMBE 2025 (remote) - Beijing (CN)
- (poster) SMBE 2024 - Puerto Vallarta (MX)
- (poster) SIBE 2024 - Naples (IT)

Main Courses and Seminars

- Statistical Course (Sarno S., Plazzi F., 2024)
- Virtual Workshop on Evolutionary Rate Covariation (Forsythe E., Sloan D.; 2024)
- Scientific Writing and Communication Course (Lövei G.; 2024)
- Inferring phylogenies from metagenomic data (Rota-Stabelli O.; 2024)
- Eukaryotic Genome Assembly Using PacBio and Hi-C (Physalia; Uliano-Silva M., Ferreira JGRN,; 2025)

Grants

Title	Institution	Date	Amount
Kurt Milton Pickett Intercontinental Award	Willi Hennig Society	2025	1,500$
Advanced Training Grant	NBFC Biodiversity Gateway	2025	530 €
Marco Polo Program	University of Bologna	2025	1,260 €
Erasmus+ Internship	European Union	2025	1,000 €

Teaching activity

- **Course design and teaching assistant at the Oslo Bioinformatics Workshop Week 2025**
  Course title: Evaluation report for smallRNA sequencing analysis, microRNA de novo annotation and expression analyses
  Duration: 15 hours
- **Tutor and co-supervisor of five studens**
  (MSc) Giuseppe Nicolella - “Caratterizzazione di sncRNA mitocondriali in *Ruditapes philippinarum* e analisi comparativa tra metazoi” (11 months; graduated 2025)
  (MSc) Alice Cascella - “Spiralia Phylogenomics” (ongoing)
  (MSc) Giovanni Galassi - “Spiralia Phylogenomics” (ongoing)
  (BSc) Giorgia Galletti - “Coevolution of Mitochondrial and Nuclear Proteins in Snakes Revealed by Network-Based ERC Analyses” (10 months; graduated 2025)
  (BSc) Nicolò Gabriele - “Analisi della Discordanza Filogenetica Nucleo-Mitocondrio nell’ordine Squamata: Evidenze da Test di Topologia” (10 months; graduated 2025)

References

Professor	Role	Institution	Contact
Federico Plazzi	PI	University of Bologna	federico.plazzi@unibo.it
Antonio M. Todaro	Collaboration	University of Modena and Reggio Emilia	antonio.todaro@unimore.it
Bastian Fromm	Host PI	Arctic University of Norway	bastian.fromm@uit.no